Team25 Multiplier Interface

A proposed document for review. Differs slightly from the multiplier we have.

Non-Pipelined Multiplier

IO

Name	Size	Input/Output	Function
clk	1	input	Multiplier clock
rst	1	input	Reset state machine
a	N	input	Multiplicand
b	N	input	Multiplier
ready	1	input	Signals to start operation
valid	1	output	Indicates output of last signaled multiplication is valid
product	2N	output	Result of operation. Only valid if valid is HIGH

Reset

Reset is synchronous and signaled by setting rst high for at least 1 clock cycle. When the reset is complete, valid will be set HIGH and product will be set to 0.

Ready/Valid Handshake

If ready is HIGH on a clock cycle, the inputs at a and b will be sent to the multiplier, regardless of whether a multiplication is in progress or not. If it is, it will be discarded as the state machine is reset.

The value product is stored in a register and is held until a new multiplication is initialized.

valid is HIGH when the data in the product register represents the result of the last initialized multiplication.

This allows for a net connection between ready and valid, which allows for a multiplication to start on the next cycle after the current one has finished. This is useful for cases where successive multiplications are needed (such as a convolution). For other cases, a controlling state machine will need to generate a single clock cycle pulse.

Timing Diagram

Google docs does not support vector images for some reason. Here is a link: https://drive.google.com/file/d/1eRjR9JceKgPTcfHCc2f1qTkfvv9jgMR8/view?usp=sharing

Pipelined Multiplier

IO

Name	Size	Input/Output	Function
clk	1	input	Multiplier clock
rst	1	input	Reset state machine and flush pipeline
ena	1	input	Enable the pipeline
a	N	input	Multiplicand
b	N	input	Multiplier
ready	1	input	Signals that inputs are good and part of a valid frame. This is shifted through pipeline stages to the output valid.
valid	1	output	Indicates that the value of the current output frame is valid.
frame	clog2(NSTAGES)	output	Indicates the number of the frame being processed next and the number of the frame that is currently at the output. It loops from 0 to NSTAGES-1.
product	2N	output	Result of operation. Only valid if valid is HIGH

Reset

Reset is synchronous and signaled by setting rst high for at least 1 clock cycle. When the reset is complete, all accumulated partial products stored in the pipeline will have a value of 0, with a valid status of LOW. frame will be set to 0,

Interface Detail

The pipelined multiplier allows for several multiplications to be conducted simultaneously with a minimal state machine. Each stage of the pipeline represents some accumulated sum of partial products.

The signal frame helps other modules capture the correct output without knowing the number of stages in the pipeline. For example, the result of the product A*B when frame=5 will be at product when frame once again equals 5.

ready and valid are not used to perform a ready valid handshake. Rather, ready serves as an indicator of if the values at the inputs a and b are to be used or discarded or not. The value at ready is shifted through each stage of the pipe line, and will show up at valid with the product of the inputs at the time. This is useful for something like a multiply and accumulate convolution, where the specific frames do not matter as much, rather that they all get added.

Timing Diagram

Google docs does not support vector images for some reason. Here is a link:
https://drive.google.com/file/d/1d6dCjTaSWY6BhlbcPrYKQ5CH5MnTjViw/view?usp=sharing